Millbrook Pump Station: Design Report

1 Design Basis

The Millbrook Pump Station serves a population of 28,093 persons. Its average dry weather flow is 3.14 MGD and its peak wet weather flow 12.9 MGD. The site layout is shown in Figure 4.

The wet well provides a working volume of 9,176 gal between the pump-off level and the lead pump's start level.

2 Pumps

Four submersible pumps are installed: P-301, P-302 and P-303 run together at peak wet weather flow, and P-304 is the standby.

Table 1. Pump schedule

Tag	Capacity (gpm)	TDH (ft)	Motor (hp)	NPSHr (ft)	Speed (rpm)
P-301	1,318	61.3	49	19.6	1,327
P-302	1,898	102.1	116	12.6	1,293
P-303	1,449	112.1	91		1,162
P-304	1,716	114.8	58	10.4	TBD

P-301, the lead pump, delivers 1,318 gpm at 61.3 ft. On the standby generator at 50 Hz, P-301 delivers 1,056 gpm.

Pump P-302 has a 71 hp motor; at runout, its capacity rises to 2,523 gpm. Pump P-304's total dynamic head is 73.2 ft.

3 Wet Well

Table 2. Wet well

Working volume (gal)	7,761
Diameter (ft)	14
Depth (ft)	29.6
High water alarm elevation (ft)	418.1

The wet well's access hatch frame is low-alloy steel (ASTM A588). The access hatch is corrosion resistant in the wet well's hydrogen sulfide atmosphere.

4 Force Main and Valves

The force main is 7,392 ft of 12-inch ductile iron pipe. Surge pressures are given in Table 7.

Table 3. Valve schedule

Tag	Size (in)	Rating (psi)	Cv	Stroke time (s)
V-11	17	268	1,689	88.2
V-12	9	150	3,317	54.6
V-13	11	152	1,916	24.5
V-14	16		2,629	15.6
V-15	19	180	2,971	22.1

Table 4. Surge analysis

Case	Pressure (psi)
Maximum transient pressure	213
Minimum transient pressure	6.9

At 1.4 mi, the force main needs no intermediate air release station. The force main is laid in high-sulfate, corrosive soil along the access road.

5 Odour Control

Table 5. Odour control unit

Fan airflow (cfm)	TBD
Carbon bed volume (ft³)	108
Fan motor (hp)	5.4

6 Electrical and Materials

The pump control panel is a NEMA 1 enclosure, mounted outdoors beside the wet well.